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The Editors
Future Generation Computer Systems
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Subject: Submission of an original research manuscript

Dear Editors,

We submit for your consideration our manuscript entitled "**CR-SHE: Oblivious homomorphic aggregation for collusion-resilient search and analytics over encrypted multi-cloud IoT data**" for publication in Future Generation Computer Systems as a regular research paper.

Internet of Things deployments increasingly replicate telemetry across several independent cloud providers for availability and vendor independence, and that telemetry is sensitive enough that it must stay encrypted once it leaves the device domain. A line of distributed-trust systems already hides which records a query selects by splitting the query across replicated servers. Our paper addresses the step those systems leave open. Selection and computation are not the same problem, and hiding the first is worthless if the second gives it back: once a scheme aggregates over the matched records, it typically hands the matched set to the providers, and the coalition recovers exactly the access pattern the selection phase protected.

We present CR-SHE, in which both keyword retrieval and analytics over the retrieved records remain oblivious to a coalition of up to n-1 colluding providers, over data encrypted at rest under a single owner's key and with no communication between providers. The central construction, oblivious homomorphic aggregation, has each provider fold its own distributed point function share directly into the homomorphically encrypted record column and return a single ciphertext, so the matched set is never assembled anywhere and its cardinality never leaves the client. A precomputed-column fast path removes the dependence on record count,

and a one-time-pad index mask, cancelled inside the ciphertext, removes the posting-list-length leakage a replicated index otherwise exposes at rest.

The work is supported by proofs of correctness and adaptive leakage security against n-1 collusion, and by an end-to-end C++ prototype evaluated on two real IoT corpora against ORAM, single-server searchable encryption, distributed-trust search and a faithful homomorphic equality scan. On 405,184 sensor records, a registered single-keyword aggregate costs 39 ms per provider irrespective of the record count, returns a fixed 526 kB response, and does 617 times less work than the homomorphic scan. A frequency attack that re-identifies every non-temporal metadata tag from an unmasked index recovers none from the masked one, and the result reproduces on a second, twelve-site corpus with three times the tag domain.

We believe the paper suits Future Generation Computer Systems because it sits squarely in the journal's scope on multi-cloud and distributed infrastructure for large-scale data: it treats replication across independent providers as a systems property to be exploited rather than merely tolerated, reports measured performance for a working prototype rather than asymptotics alone, and states plainly what the evaluation does not establish.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have read and approved the submitted version and agree to its submission. The authors declare no competing financial interests or personal relationships that could have appeared to influence this work. The manuscript has not been posted to a preprint server.

Thank you for considering our submission. We look forward to the reviewers' comments.

Yours sincerely,

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